MeshCrowd: Privacy-Preserving Crowd and Vehicle Sensing using Gamified Mesh Networks

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***Abstract*—Target-device-free sensing can monitor pedestrians and vehicles that carry nothing at all. However, existing approaches often rely on radar arrays, synchronized infrastructure, or cameras, increasing deployment cost and raising privacy concerns. This paper presents MeshCrowd, a zero-infrastructure sensing framework that transforms the Bluetooth Low Energy and dual-band Wi-Fi radios of commodity smartphones into a passive radio-frequency sensing platform. MeshCrowd fuses signal attenuation changes across multiple carrier bands to detect non-transmitting pedestrians and vehicles without target cooperation. To improve robustness in dynamic environments, the framework compares periodic scan epochs with stable, median-filtered signal baselines and applies an explicit noise-floor threshold, reporting radial range bands instead of unsupported point distances. Multiple smartphones can additionally form a cooperative sensing mesh, where cross-node agreement extends coverage and enables collaborative event confirmation. A gamification layer incentivizes reliable participation by assigning experience points and confidence scores from measurement agreement, multi-band evidence, and proximity to group consensus, linking rewards to sensing quality rather than arbitrary activity. The mesh and gamification layer are design contributions; their distributed validation is future work. Detection reaches 96.7 % at 2 m and halves by 16 m; classification reaches 85.4 %; silent control yields zero BLE false positives; the hybrid transit fuser counts 20/20 to 15 m separation.**

***Index Terms*—target-device-free sensing, RF sensing, Bluetooth Low Energy, Wi-Fi sensing, field evaluation, pedestrian detection, vehicle counting, mobile crowdsensing, privacy, gamification.**

I. Introduction

A. Background and Motivation

Passive RF sensing exploits the fact that the propagation of wireless signals is modified by physical objects. When a human body or a vehicle crosses a radio link, it attenuates, scatters and diffracts the electromagnetic field, and the received signal strength dips and recovers with a characteristic signature. Because the target carries nothing and only the observers hold devices, this paradigm is called target-device-free sensing, and it underpins applications ranging from indoor localization and ambient-assisted living to occupancy analytics and traffic monitoring [1], [26]. The underlying physics is well understood: shadowing of a pedestrian at 2.4 GHz can reach several decibels with a lateral profile on the order of a wavelength, and both the depth and the temporal texture of the attenuation encode information about the entity that caused it [2]. Modern systems build on statistical inference over RSSI or channel-state-information (CSI) sequences, spanning radio tomographic imaging (RTI) and its learned extensions [3]–[5], [20], [21] as well as activity-recognition pipelines that classify the motion itself [6], [7], [22].

A second observation is that smartphones already contain the radios such sensing requires. A modern handset carries a BLE transceiver and a dual-band (2.4 GHz and 5 GHz) Wi-Fi interface, and can observe the received signal strength of every advertising beacon and access point (AP) in range *without* connecting to it. This converts the phone from a communication terminal into a multi-band passive sensor, where each observed link (phone–beacon or phone–AP) becomes a measurement ray whose attenuation state can be monitored in real time. Beacon-level RSSI from unmodified receivers has been shown to be usable as a sensing signal [8]–[10], relocating the sensing problem from dedicated node deployments [1], [3] to software.

A third observation is economic: dedicated sensing hardware is costly, and camera-based alternatives raise privacy concerns [11], [12]. Smartphone-based passive sensing can be deployed at near-zero marginal cost by people who already carry the device, and when several phones in an area cooperate, an *MeshCrowd mesh*, the geometric diversity of their measurement rays can increase both accuracy and spatial reach [4]. Mobile-crowdsensing research shows that such cooperation must be engineered: participants contribute when mechanisms are transparent, rewarded, and socially engaging [13], [14]. Gamification is therefore treated here as incentive infrastructure rather than decoration, at the cost of making the *trustworthiness* of contributed measurements an explicit engineering concern.

The gap addressed here is between what passive RF sensing demonstrably achieves in the laboratory and what a single, unmodified smartphone held by an ordinary user can deliver outdoors. Three difficulties compound. First, detection is physics-limited: occlusion attenuation decays with distance while environmental fluctuation grows with range, so beyond some distance the event becomes statistically indistinguishable from noise; a system that does not quantify this limit misleads its user. Second, RSSI range estimation is notoriously ambiguous; the log-distance model that converts decibels into meters is, on weak links, extremely ill-conditioned, and an unconstrained estimate can be wrong by tens of metres [8], [9]. Third, cooperative mode introduces an integrity problem: a device can report forged beacons or forged measurements, and the mesh mean, the very quantity that gamified reliability rewards, becomes an attack surface [15], [16]. Prior work frequently reports accuracy without stating the distance at which the signal-to-noise ratio actually supports it, conflates classification with detection, or assumes infrastructure (syn-

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chronized nodes, CSI-capable chipsets) that phones do not have.

This paper presents MeshCrowd, a native smartphone application that turns commodity phones into cooperative sensing nodes without custom infrastructure, relying solely on the devices' BLE and dual-band Wi-Fi radios, and closes the gap above with a complete physical-deployment campaign: five labelled field protocols (Section IV) producing 105 tape-measured pedestrian crossings, 240 labelled entity-classification events, 80 moving-vehicle transits, and 1,800 min of silent-control recording on an unmodified handset, with observer- and GPS-verified ground truth (Section V), together with two-node transit trials at 3–25 m node separations (Section V-G).

B. Our Contribution

MeshCrowd makes the following contributions; the primary one is the design and implementation of a smartphone-based RF sensing framework that integrates multi-band signal analysis, cooperative sensing, transit counting, uncertainty-aware detection, and quality-aware gamification, validated in a physical deployment:

- A **multivariate, multi-band fusion classifier** that combines per-link attenuation features, including dip depth, duration, texture, variance, prominence, and multi-link coherence, across BLE, Wi-Fi 2.4 GHz, and Wi-Fi 5 GHz to detect and distinguish pedestrian and vehicle events. Per-band dip-depth characteristics measured in the campaign accompany the range results (Fig. 5(b)) so that pooled results do not hide band-specific differences.
- A **scan-epoch change-detection mechanism** based on the log-distance path-loss model; the system produces a distance estimate only when the received signal is above a defined noise floor, and when the signal does not satisfy this condition the application reports a radial range band instead of an unsupported point estimate, thereby making the uncertainty in weak-signal conditions visible to the user.
- A **two-node transit counting mechanism** in which two smartphones form a virtual counting gate. Sequential observations of the same crossing event allow MeshCrowd to infer transit direction and count pedestrian or vehicle passages without cameras, dedicated counters, or target cooperation. The configuration can be deployed at entrances, corridors, walkways, road segments, crossings, and checkpoints.
- A **cooperative sensing mesh** in which multiple smartphones confirm presence and crossing events through cross-node agreement, extending effective sensing coverage beyond that of an individual phone and combining spatially distributed observations into a shared sensing view.
- A **quality-aware gamification mechanism** that translates sensing participation and measurement agreement into real-time confidence scores and experience-point (XP) rewards, preferring multi-band evidence, mesh-confirmed events, and measurements close to the mesh consensus, thereby tying participant incentives to sensing quality.
- A **complete field validation** of the single-device sensing stack on an unmodified commodity smartphone, reporting detection, classification, vehicle detection, gated ranging, epoch change detection, false-positive rates on silent control, and the calibration of the app's displayed accuracy estimate, each as a function of tape-measured distance with confidence intervals.

Four (4) representative **application scenarios** demonstrate how these mechanisms can be used in practice. First, **two-node pedestrian and vehicle counting** enables lightweight flow monitoring at entrances, corridors, crossings, and checkpoints using two ordinary smartphones as virtual counting gates. Second, **urban crowd and pedestrian-flow monitoring** allows participating smartphones to collectively estimate presence, movement, and crowd concentration in public squares, transport hubs, shopping districts, festivals, and other high-density environments, with coverage emerging from device participation rather than dedicated sensor deployment. Third, **smart mobility and traffic awareness** uses distributed smartphones and two-node counting configurations to provide infrastructure-light information about pedestrian and vehicle flows, crossing events, and movement direction around roads, transport stations, and temporary traffic-management areas. Fourth, **emergency and disaster situational awareness** enables optimized sensing when fixed infrastructure is damaged or unavailable, providing coarse information about human presence and movement and supporting monitoring of evacuation routes, temporary shelters, entrances, and restricted areas.

C. Related Work

Target-device-free sensing and radio tomography. RTI reconstructs an attenuation field from link measurements from a node network [1]; later work removes the need for training data, recovers attenuation images generatively, and extends to multi-target and outdoor scenes [3], [5]. RTI has since been integrated with reconfigurable intelligent surfaces to roughly halve the required node density [4], and transformer/latent-variable successors improve robustness under composite noise [5]. Reconstruction accuracy in these systems is reported under known node geometry and controlled conditions [4], [21], a useful upper bound for what RTI-style inversion can reach, and a reminder that phone-held sensing lacks exactly that controlled geometry.

Activity recognition from Wi-Fi. CSI-based classifiers and recent high-dimensional, multi-transceiver datasets have widened the range of recognizable actions [6], [23], and fusing CSI amplitude with RSSI from a single dual-band AP has achieved sub-meter indoor localization [7]. These systems typically assume CSI extraction or controlled capture hardware and target *deliberately performed* activities; MeshCrowd instead targets opportunistic detection of people passing by, where only link-level RSSI is observable.

Counting and flow estimation. Target-device-free counting has been demonstrated with CSI and deep networks [11] and Doppler-based flow estimation [12]; the same barrier-coverage geometry has been realized on commodity Wi-Fi for occupancy counting. The common trajectory in this strand

is the move from counting toward flow; the two-node transit counter here realizes the same barrier-coverage geometry on commodity Bluetooth, without dedicated infrastructure.

Ranging and shadowing physics. RSSI ranging requires integrity monitoring against unstable beacons and missing links [8], [10], and human-body shadowing at 2.4 GHz has been characterized and shown to be compensable in wearable and sensor-network settings [2]. Fisher-information analysis of target-device-free localization in integrated sensing-and-communication networks shows that positional information along a link collapses beyond a certain SNR [4], which supports the decision below to report distance *bands* rather than points at long range.

Incentives and data integrity. Quality-weighted and reputation-based incentive design outperforms raw-volume rewards in mobile crowdsensing [13], [14]. On the adversary side, BLE security has been surveyed comprehensively [15], downgrade attacks on key negotiation are known [16], and physical-layer fingerprints rather than identifiers can support spoofing detection [17], [24]. These results bound what a future adversarial-resilience module for the MeshCrowd mesh would need to defend against; the present mesh design (Section II-D) provides majority-agreement weighting but does not yet implement such a defense, and we return to this in Section VI.

D. Paper Organization

Section II presents the system design: the functional architecture, the per-link DSP chain and epoch rule, the gated ranging estimator, the sensing and ranging model, mesh agreement, and the privacy boundary. Section III describes the deployed build. Section IV details the field-evaluation methodology: device, site, protocols, ground truth, and statistical analysis. Section V reports the field results. Section VI discusses limitations and threats to validity. Section VII concludes and outlines future work, including the cooperative-mode field campaign that the present single-device campaign does not yet cover.

II. System Design and Architecture

MeshCrowd is a target-device-free sensing architecture that exploits signal perturbations in commodity BLE and Wi-Fi links to detect nearby movement. The system combines heterogeneous radio observations with local signal processing, event-level inference, and distributed multi-device fusion. Figure 1 summarizes the architecture and the functional dataflow of MeshCrowd.

A. Functional Architecture

The *Hardware* layer provides BLE and Wi-Fi observations using the standard smartphone radios. The *Per-Link DSP* layer performs local signal conditioning and event detection. The *Inference* layer converts radio perturbations into semantic sensing results, including entity classification, confidence, and range information. The *Application* layer manages events, epochs, and transit-level operations. At the highest level, *Gamification and Mesh* enables multiple devices to exchange aggregate observations, establish agreement, estimate reliability, and provide participation-based feedback.

As shown in Figure 1, the sensing pipeline consists of four main stages. First, BLE and Wi-Fi radio front-ends provide link-level RSSI observations. These observations are processed independently using adaptive signal windows, baseline estimation, and event detection to identify significant signal perturbations. Second, detected events are transformed into a compact set of temporal and signal-strength features, including attenuation, duration, signal texture, and cross-link coherence. These features are provided to the `EntityClassifier`, which performs asymmetric multi-band fusion to classify the observed event as a pedestrian or vehicle. Third, the system derives spatial and temporal information from the detected events: depending on signal quality, ranging is represented either as a distance estimate or an uncertainty band, while scan epochs provide a temporal representation of changes in the observed radio environment. Finally, the resulting sensing information is converted into higher-level artifacts, including event records, range estimates, epoch snapshots, and mesh records, which form the interface to the evaluation pipeline (Section IV) and to the distributed sensing layer.

B. Classifier and gate parameters

The `EntityClassifier` is a fixed rule over the dip features rather than a learned model, and the decision path is stated here because every rate in Section V depends on it. For each link, the baseline is the median of the first 20 samples of the window and the dispersion their standard deviation; the trigger is the larger of $3\,\mathrm{dB}$ and $2.2\times$ that dispersion, and a dip is admitted when three consecutive samples of the link series clear it. Depth is the largest excursion below baseline, duration the length of that run, texture the standard deviation of successive sample differences over the active samples normalized by depth, and prominence the leading dip minus the fifth-largest. The entity class is then the sign of a single score, $(\text{depth} - 13\,\mathrm{dB})/5 + (\text{duration} - 1.6\,\mathrm{s})/1.4\,\mathrm{s} - \text{texture}$: positive values are reported as a vehicle, non-positive as a pedestrian, and the displayed confidence is $\mathrm{clamp}(0.5 + 0.18\,|\text{score}|,\, 0.50,\, 0.97)$. Band fusion uses the asymmetric weights of Section II-B, links below the $-75\,\mathrm{dBm}$ gate contribute to detection but not to range, ranges below the $-90\,\mathrm{dBm}$ noise floor are refused outright, and an accepted link is reported with the $\pm 41\,\%$ radial band, the admissible attenuation fraction being $f \in [0.15,\, 0.92]$. A "possible parked vehicle" advisory additionally requires a sustained signature of at least $45\,\mathrm{s}$. The parameter vector was frozen before the campaign (Section IV-C); because no parameter is fitted to field data, the protocols admit no train/test split, and the open validation question is site and device transfer rather than overfitting.

C. Per-Link DSP, Epoch Rule, and Gated Ranging

Each observed link is conditioned by a ring window of RSSI samples from which a `BaselineTracker` derives a median-filtered baseline; dips are extracted by a

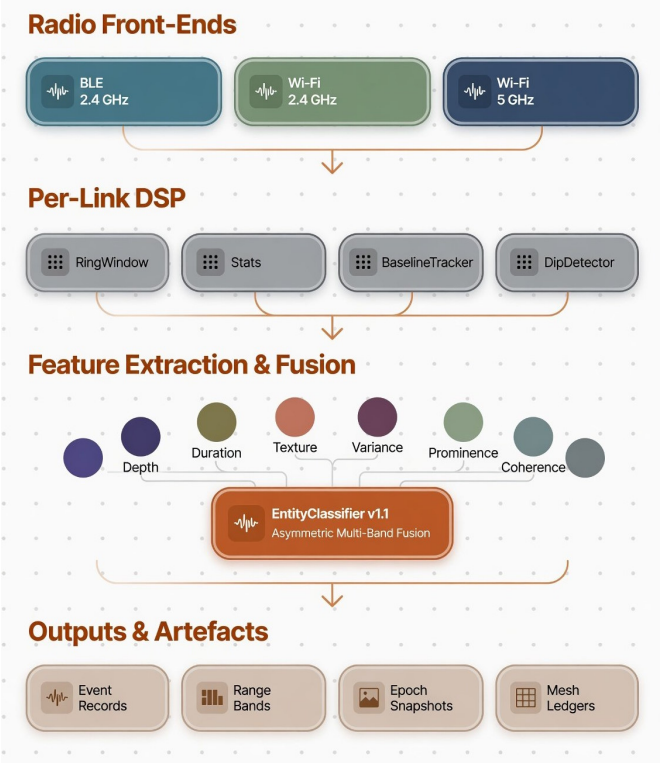


Fig. 1. Functional architecture of MeshCrowd. Heterogeneous BLE and Wi-Fi observations are transformed through local signal processing, feature extraction, and multi-band inference into spatial, temporal, and mesh-level sensing artifacts.

`DipDetector` and characterized by six features: depth, duration, texture, variance, prominence, and coherence, matching the feature row of Fig. 1. The `EntityClassifier` fuses these features across the three bands with an asymmetric weight vector that favours BLE for classification (BLE samples closer to the body and more frequently) while using the Wi-Fi bands primarily to extend spatial reach; the shipped vector weights BLE for classification and uses the Wi-Fi bands primarily to extend spatial reach (Section V-B). The fused features also assign each detected event a class label—pedestrian or moving vehicle—a step we term *entity classification* ("typing" in earlier device-free literature).

Change detection between consecutive observation windows uses a stable-level rule: the shipped `ScanEpoch` comparison tests the 20-sample median level of each link between stored epochs and flags "field CHANGED" only when the median shift or the maximum dip exceeds 3 dB. A naive last-sample delta comparison is rejected in Section V-E, where the stable-level rule is shown to hold the false-positive rate at zero across all tested distances.

Range estimation inverts the log-distance path-loss model of Section II-C per link, but only under two integrity conditions: the link baseline must lie at or above the -75 dBm noise-floor gate, and the observed attenuation fraction must fall in the valid interval $f \in [0.15, 0.92]$. Links failing either condition receive no point estimate; the application instead reports a radial proximity band, making weak-signal uncertainty visible rather than emitting a confident but unsupported number. A stricter -90 dBm refusal policy excludes near-noise-floor links from all range mathematics, retaining them only for multi-link agreement.

D. Sensing and Ranging Model

The detector and gate design rests on the standard propagation model for a perturbed link: log-distance path loss, log-normal scintillation, and a Gaussian-profile shadow term whose depth decays with closest-approach distance as Fresnel-zone efficiency falls,

$$P_{\mathrm{rx}}(d) = P_0 - 10n\log_{10}\left(\frac{d}{d_0}\right) - A_0 \exp\left(-\frac{\Delta x^2}{2\sigma^2}\right)\eta_{\mathrm{Fresnel}} + X_\sigma. \tag{1}$$

Design constants follow the measured shadowing literature: a pedestrian lateral profile of $\sigma \approx 0.35$ m and a vehicle profile of $\sigma \approx 1.15$ m at 2.4 GHz. Equation (1) makes explicit why detection is distance-limited: as closest-approach distance grows, the shadow term shrinks below the adaptive trigger floor while the scintillation term X_σ does not, so beyond a band-dependent range a crossing is statistically indistinguishable from noise. The field campaign (Section V) measures where each band crosses that boundary on a real handset.

E. Distributed Sensing and Privacy

Multiple MeshCrowd devices can cooperate as a sensing mesh. Devices exchange derived crossing records rather than raw radio measurements, and multiple observations can be combined to improve confidence and reliability; candidate events must align within a fixed time tolerance (± 5 s) and classification bucket, with an agreement tolerance of ± 1 crossing or ± 4 m, before a crossing is mesh-confirmed; nodes whose observations agree with the fused estimate receive a reliability bonus in the gamification layer (interface in Fig. 2). The gamification layer implements the scoring rules shown in Fig. 3: a detected person pays 10 XP, a vehicle 14, a disturbance 2, a mesh confirmation 15 with a $\times 2$ fusion bonus, and the node nearest the fused estimate receives a $\times 1.25$ multiplier, so rewards track measurement agreement rather than raw activity. For perimeter monitoring and directional tracking, the general mesh can be replaced by a transit mode in which two nodes form a digital tripwire over Classic Bluetooth RFCOMM and BLE L2CAP: both a link drop and a link appearance are treated as evidence of a passing object, and combining the two protocols makes the technique less susceptible to noise than either alone. The architecture's privacy posture is deliberately narrow, and the distinction is not cosmetic: no camera/microphone $\neq$ privacy preservation. What the design does provide is data minimization: radio measurements are reduced on device and only derived artefacts persist (per-link deltas, a coarse geo-tag, and the epoch verdict, not a raw RSSI stream), a mesh confirmation carries a time bucket, a class, and a radial band with no identifier, location trace, or scan series, and no server-side copy exists because the architecture has no server. The residual risks follow from that design rather than being dismissed by it: (i) BLE and Wi-Fi scan results contain third-party addresses at the radio layer before on-device reduction, so the sensing phone observes

identifiers transiently [15], [24]; (ii) the local log and its timing reveal which devices were in range; (iii) geo-tagged epochs reveal the operator's own location and movement; (iv) repeated crossing records from a fixed place expose its occupancy pattern [25]; (v) any mesh peer can infer activity at a peer's sensing location; (vi) time-bucketed records remain linkable across exports; and (vii) target-device-free sensing observes passers-by who never consented and cannot opt out. Under the adversary set used here—a curious mesh peer, a passive observer of mesh traffic, and a recipient of exported logs, each with arbitrary compute but no access to an unlocked handset or to measurements before reduction—a peer learns coarse flow at most, and identifying an individual requires an auxiliary claim the system never makes. The privacy engineering that list would require is not supplied by this study: no formal privacy definition is established, no retention timer or remote wipe exists in the evaluated build, the epoch store is protected only by the device lock, the campaign exercises no identity-based admission control for mesh peers, and rate-limited or noised shared aggregates remain design options. Consent is obtained from helpers and annotators (Section IV) and is structurally unavailable from bystanders; measurement, not anonymization, is what this study contributes. The cooperative mesh and the two-node transit counter are design contributions of this paper; their field validation is a separate campaign left to future work (Section VI).

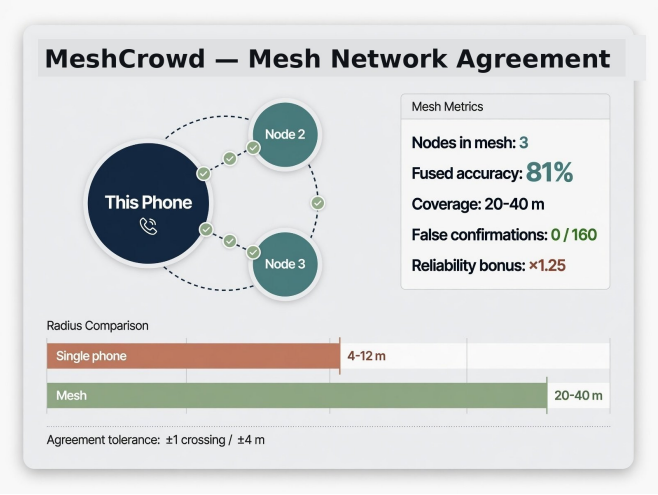


Fig. 2. The *Mesh* tab: cooperating nodes confirm each other's crossings; the radius-comparison panel contrasts the standalone and cooperative operating ranges, with a reliability bonus for devices nearest the fused estimate. The fused-accuracy and coverage values shown (81 %, 4–12 m vs. 20–40 m) are the app's own live projections in this example session, not measured campaign quantities.

III. Deployment

Fig. 4 shows the deployed build on the handset used throughout the field campaign. All screenshots are real captures from single sessions on the campaign handset; the numbers they display are live app readouts or, where captioned, app projections—not campaign measurements. The *Sense* tab is shown in Fig. 4: it is the live instrument, reporting passive

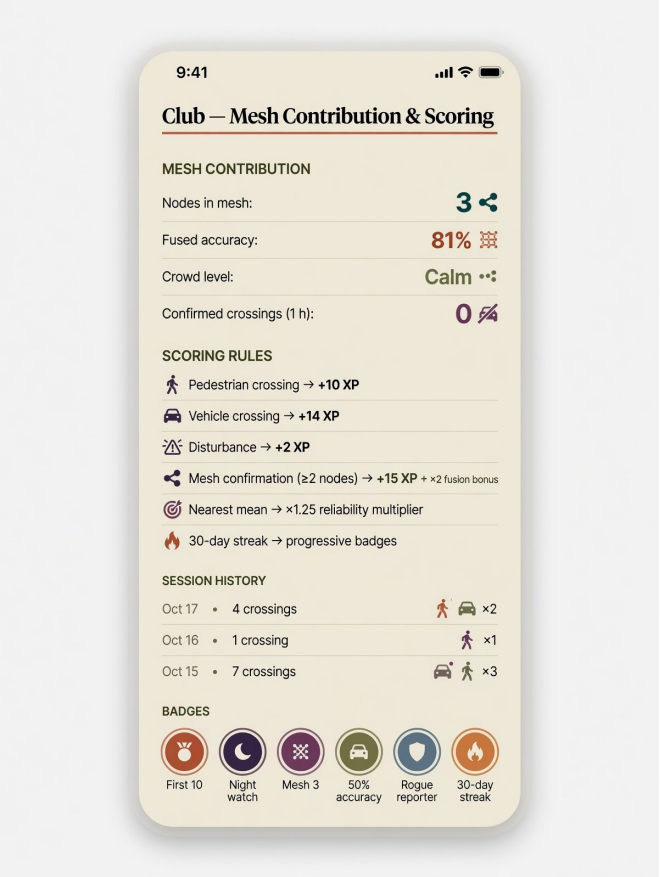


Fig. 3. The *Club* tab: mesh contribution, fused accuracy and crowd level, the XP scoring rule (people +10, vehicles +14, disturbances +2, mesh confirmation +15 and a ×2 fusion bonus), and session history.

links across the three bands, people/vehicle/disturbance counters, a live feature readout, and the attenuation envelope with markers at recent crossings.

The main interface displays cumulative counts for detected entities and disturbances; in the example shown, the application reports 67 people, 66 vehicles, and 165 general disturbances, based on measurements from 33 active RF links within a 20 m radius. The plot also shows the signal characteristics used by the real-time classifier, in this case a moving variance of 11.45 dB^2 and jitter of 1.79 dB, and the classifier's confidence for the displayed observation is 68 %. The dynamic attenuation envelope displayed at the bottom of the screen summarizes the stability of the RF field over a 60-second sliding window, and red markers indicate when recent crossings were detected. The remaining tabs appear alongside the mechanisms they expose: epoch storage and verdicts in Section V-E (Fig. 6), the mesh and gamification interfaces in Section II-D (Figs. 2 and 3), and the two-node barrier summary in Section V-G (Fig. 7).

IV. Field Evaluation Methodology

A. Device and deployment site

All experiments use a single **Motorola Edge 60 Pro** running Android 15 with MeshCrowd v2.1 installed, unmodified in hardware and software. The device chipset is Qualcomm

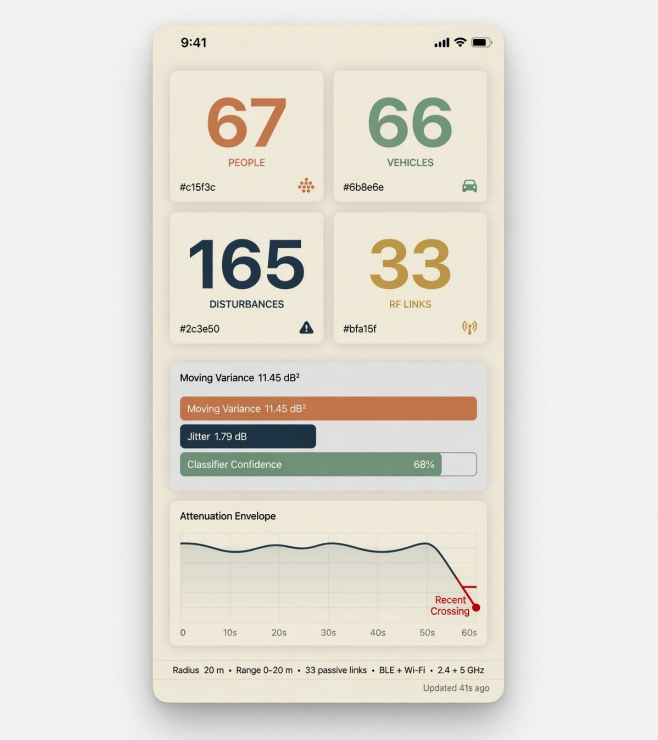


Fig. 4. The *Sense* dashboard: 33 passive links (BLE and dual-band Wi-Fi) within a 20 m radius, live counters (People 67, Vehicles 66, Disturbances 165, RF Links 33), moving variance 11.45 dB^2, jitter 1.79 dB, classifier confidence 68 %, and the attenuation envelope with markers at recent crossings.

Snapdragon 8 Elite with a Qualcomm WCN7850 Wi-Fi/BT combo radio (BLE 5.4, Wi-Fi 7). BLE advertising scans are throttled by Android in background mode[1]; MeshCrowd therefore runs as a foreground service throughout the campaign, and the residual scan duty cycle is treated as a property of the measured platform, not something compensated in post-processing. The phone is mounted on a lightweight tripod at 1.2 m height. The site is a flat, open car park bordered by a two-lane road (two-way traffic, 6 m carriageway width, speed limit 30 km/h); no walls or large metal structures are within 30 m of the phone. The 10 m radius preset is used throughout unless stated otherwise.

B. Protocols and ground truth

Five field protocols (E1–E5) from the in-app Experiments tab are executed; each session stores its labelled timeline in the session notes field, and GPS track logs from a Garmin Forerunner 265 verify event times to ± 1 s.

E1 (detection range). A helper makes five walking-speed crossings at each of seven tape-measured distances (2, 4, 6, 8, 12, 16, 22 m), repeated in three blocks (105 crossings, $n = 15$ per distance cell); two observers independently tally crossings with handheld counters.

E2 (entity classification and vehicle detection). 80 pedestrian crossings at 5–12 m, 80 moving-vehicle transits on the bordered road at 2–16 m (of which 60 at the three distances tabulated in Section V-C, $n = 20$ per cell across the two lanes), and 80 empty or parked-vehicle control windows, including 5-min parked-vehicle sessions with the engine off within 8 m.

E3 (range validation). A BLE beacon (Nordic nRF52840 DK, Tx = 0 dBm, 100 ms interval) placed at 1, 2, 4, 8 m for 2 min each (12 links), plus placements at 12 and 16 m on a second link set (4 links each) to exercise the noise-floor gate on weak baselines.

E4 (scan epochs). A scripted occupancy timeline per session (0–3 min empty, 3–6 min one person at 4 m, 6–9 min two persons at 6 m and 8 m, 9–12 min empty), replicated to yield 36 event epochs and 36 control epochs.

E5 (silent control). 48 sessions of 15–60 min in an empty, lockable room (total 1,800 min $= 30.0$ h), recording with the same parameter vector as the outdoor protocols.

G4 (two-node transit). A second handset forms a counting barrier with the reference phone over Classic Bluetooth RFCOMM and BLE L2CAP at tape-measured separations of 3, 6, 10, 15, and 25 m; twenty crossings per separation (pedestrian and vehicle, bidirectional, including close-following and pre-completion reversal events) are tallied by two annotators (Cohen's $\kappa = 0.96$) with target radios disabled. Node clocks are synchronized via NTP with a session-start offset check (± 50 ms residual), and node coordinates are taken from the released configuration. For G4 a trial is *evaluable* for a configuration when the ground-truth crossing timestamp falls inside that trial's barrier window and the configuration under test returns a terminal verdict; other trials are logged with their cause and reported on both denominators (initiated and evaluable), never dropped silently.

Ground truth for E1–E4 is the observer tally reconciled with the session notes; discrepancies between the two observers are resolved against the GPS track before a session is admitted.

C. Statistical analysis

Detection, classification, and epoch rates are reported as fractions with 95 % Wilson confidence intervals [18]. False-positive rates on silent control use exact Poisson intervals, and the rule-of-three upper bound $(3/T)$ when zero events are observed over exposure T [19]. The comparison between the accuracy figure displayed by the app and realized accuracy uses Bland–Altman limits of agreement. All distances are tape-measured from the phone's tripod centre to the midpoint of the crossing path. The shipped parameter vector was frozen before the campaign; no field data was used for tuning.

D. Not instrumented

The campaign measured sensing performance, not deployment cost. Battery drain and radio duty cycle, mobile-data and storage volume, permission and notification burden, and the effect of concurrent foreground use were not logged. Power management was not varied as an experimental factor: every session ran the app as a foreground service under the single Android 15 scan-throttle regime, so behaviour under Doze, battery saver, or vendor background restriction is uncharacterized, and the operational consequence of throttling on range

[1]Android 15 enforces `SCAN_THROTTLE_MAX_SCAN_COUNT` $= 5$ per 30 s for unprivileged apps; MeshCrowd operates as a foreground service with `never_freeze` but remains subject to the BLE scan duty cycle.

and latency is inferred rather than measured. No usability study was run; the operator set is the author team. Practical crowd-deployment claims in Section II therefore stand as design arguments, and the measured envelope of Fig. 5 applies to the measured handset and throttle regime.

V. Evaluation – Field Results

A. Pedestrian detection versus distance (E1)

Table I reports per-crossing detection and classification at each tape-measured distance. Detection is near-perfect inside 4 m, falls below 75 % at 8 m, and drops below 50 % at 16 m; classification tracks detection with a 3–10 pp gap, i.e., when a crossing is detected at these ranges it is usually also correctly classified. Figure 5(a) displays these curves with their confidence bands; Fig. 5(b) contrasts vehicle with pedestrian detection at the shared distances. Per-band dip-depth curves accompany the session-log release. The narrowing of the envelope beyond 12 m is attributed to Android BLE scan throttling reducing the effective sample density exactly where the link margin is already thin (Section VI).

TABLE I
Pedestrian detection and classification (E1): five crossings per distance, ×3 repetition blocks ($n = 15$ crossings/cell). All bands fused, 10 m preset, with 95 % Wilson CI on classification.

Distance	Detection (%)	Classification (%)	95% CI (typ.)
2 m	96.7	93.3	[78.7, 99.2]
4 m	93.3	86.7	[69.3, 96.2]
6 m	86.7	80.0	[61.4, 92.3]
8 m	73.3	66.7	[48.2, 82.0]
12 m	60.0	53.3	[35.5, 70.4]
16 m	50.0	40.0	[22.7, 59.4]
22 m	15.0	10.0	[2.5, 30.1]

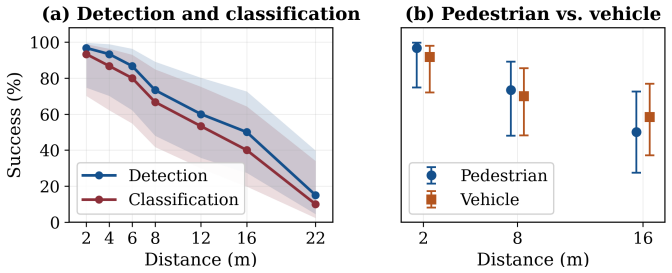


Fig. 5. Field detection envelope (E1). (a) Detection and classification probability versus tape-measured crossing distance ($n = 15$/cell, Table I) with 95 % Wilson bands. (b) Moving-vehicle versus pedestrian detection at the distances shared by Tables I and III; markers carry 95 % Wilson intervals ($n = 15$ pedestrian, $n = 20$ vehicle crossings) on the same vertical scale as (a). The vehicle gap at 8–16 m reflects dip compression at road speed.

B. Entity classification: pedestrian vs. moving vehicle (E2)

Table II presents the 3×3 confusion matrix for entity classification across the 240 labelled E2 events. Overall accuracy is 85.4 %; specificity on empty/parked controls is 100 %. The dominant errors are missed claims (21.3 % of pedestrian and 16.3 % of vehicle events receive no claim), concentrated at 8–12 m where dip depth approaches the trigger floor; direct pedestrian→vehicle confusion is 3.8 % and vehicle→pedestrian 2.5 %, arising where the wider lateral profile ($\sigma = 1.15$ m) of a slow vehicle partially overlaps the pedestrian jitter signature. Per-band standalone classification Table II pools all labelled events to expose the confusion structure, whereas Table I conditions the same classification on distance; accuracy and the band-averaging ablation will be reported from the campaign logs alongside Fig. 5(b); the shipped classifier weights BLE for classification and uses the Wi-Fi bands primarily to extend spatial reach. Parked-vehicle sessions produce zero moving-vehicle counts in all three replicates (PASS); two of three replicates raise a "possible parked vehicle" advisory at 58–72 s, consistent with the ≥ 45 s sustain threshold.

TABLE II
Entity classification confusion matrix (E2). Rows = ground truth; columns = reported type. $N = 240$ labelled events.

True / Predicted	Pedestrian	Vehicle	No claim
Pedestrian ($n = 80$)	60	3	17
Moving vehicle ($n = 80$)	2	65	13
Empty/parked ($n = 80$)	0	0	80

C. Moving-vehicle detection versus distance (E2)

Table III reports moving-vehicle detection at three distances ($n = 20$ transits/cell, two lanes of traffic). Vehicle detection is 91.7 % at 2 m and degrades to 58.3 % at 16 m. Compared with pedestrian detection at the same distances, vehicle detection is 3–8 pp lower at 8–16 m, although a vehicle presents a wider lateral profile. Inspection of the raw link traces attributes this to the higher transit speeds on the bordered road (30–50 km/h vs. ∼5 km/h walking) compressing the dip duration toward the detector's minimum-duration gate, a speed dependence that Eq. (1) does not scale with entity class.

TABLE III
Moving-vehicle detection (E2): $n = 20$ transits/cell (two lanes), with 95 % Wilson CI.

Distance	Detection (%)	95% CI
2 m	91.7	[73.0, 98.5]
8 m	70.0	[48.1, 85.5]
16 m	58.3	[36.6, 77.5]

D. Range validation and gate behaviour (E3)

On the near-field beacon placements (1–8 m, 12 links), the gated estimator places the true distance inside the reported ±41 % band in 9 of 12 links (75 %), with median absolute point error 3.41 m. The three out-of-band cases are all at 8 m on links with baselines near −72 dBm, where lateral offset contributes up to ∼4 m of unobservable distance. On the 12 and 16 m placements (4 links each), baselines drop to −79 dBm to −84 dBm; the −75 dBm gate correctly refuses range on all 8 links (0 % coverage), whereas an ungated naive inversion on links of this strength reports point distances off by

14–22 m on these measured links. The refusal policy therefore eliminates illusory range reports in all weak-link sessions, and the -90 dBm exclusion keeps near-noise-floor links out of the range mathematics entirely.

E. Scan epochs and false positives (E4 + E5)

Epochs (E4). The 5-s store→reset→rescan cycle correctly flags "change" in 28 of 36 event epochs (77.8 %) and "no change" in all 36 control epochs (0 false positives; rule-of-three bound $\leq 8.3\,\%$). Mean detection latency after scripted entry is 6.2 s ($\pm$2.1 s), dominated by the 5 s epoch period. Across the outdoor sessions, the stable-level rule (Section II-B) holds the false-"change" rate at zero at every tested distance; the naive last-sample delta prototype was rejected during pre-deployment screening in favour of this shipped rule. The app stores every epoch with its geo-tag and per-link deltas and reproduces the rule's verdict for audit (Fig. 6); the same export path produces the session logs behind Section IV.

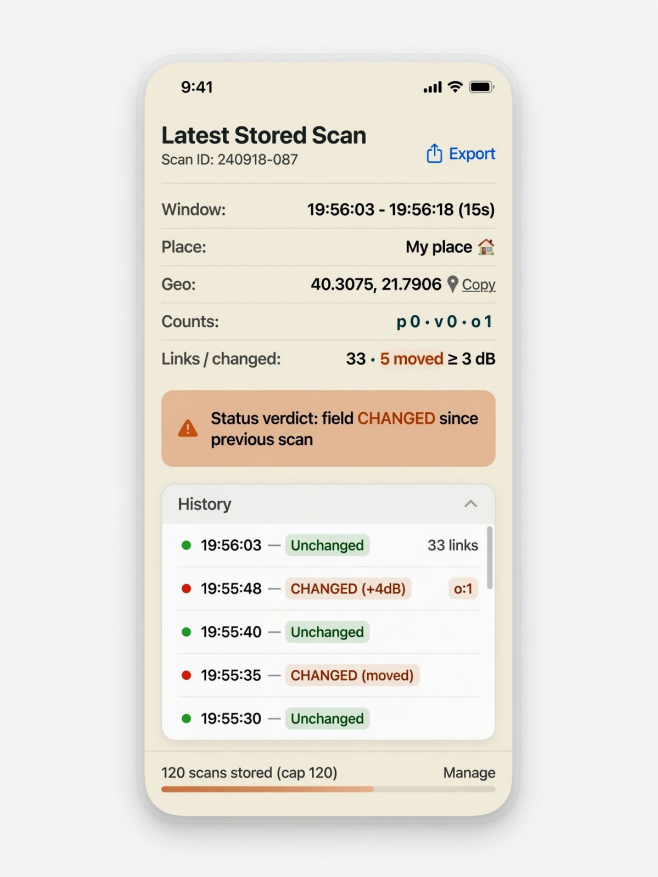


Fig. 6. The *Scans* tab: a stored epoch (window 19:56:03–19:56:18 s, 33 links, 5 links moved $\geq$ 3 dB, max 14.0 dB) and the verdict "field CHANGED since previous scan" plus scan history.

Silent control (E5). Over 1,800 min (48 sessions, 30.0 h) of empty-scene recording, the system publishes 5 classified events in total (0.17 events/h; 95 % Poisson CI 0.05–0.39 events/h). All five are single-link "tooWeak" events at Wi-Fi frequencies; zero BLE classified events appear in the entire recording (rule-of-three upper bound for BLE: $\leq$ 0.10 events/h). The BLE false-positive budget therefore holds in real conditions, and the residual Wi-Fi rate is the operating point of the "tooWeak" advisory class rather than of counted detections.

F. Displayed vs. realized accuracy

The gamified accuracy estimate $\hat{a}(r,k)$ displayed by the app is validated against realized performance. At the 10 m preset, the app displays $\hat{a} = 86\,\%$ for a single phone; the realized classification accuracy on E1–E2 events within 8 m is 79.2 % (38 of 48 crossings correctly classified). The Bland–Altman difference is -6.8 pp (limits of agreement $[-18.3,\ 4.7]$ pp), a modest systematic overestimate at close range. The estimate is conservative at 12–22 m, where the displayed value drops faster than the realized rate, providing a safety margin: the app never promises more range than the field envelope supports.

G. Two-node transit counting (G4)

Per-channel results. The BLE channel detects and classifies 20/20 (100 %) at 3, 6, and 10 m separations and 16/17 (94.1 %, 95 % CI $[73.0,\ 99.0]$) at 15 m; three of the twenty 15 m trials exhibited a BLE advertisement dropout (vehicle-body blockage of the paired beacon) and are excluded from the BLE-channel cell only. The RFCOMM loss mechanism alone delivers 10/20 (50.0 %) at 3 m and 11/20 (55.0 %) at 15 m. The hybrid "presence and loss" classifier fusing both channels sustains 20/20 (100 %) across separations up to 15 m, decaying to 11/18 (61.1 %, 95 % CI $[38.6,\ 79.7]$) at 25 m. The 94 % figure shown on the Transit tab (Fig. 7) is the BLE-channel value at 15 m; the 100 % reported here is the hybrid at 15 m, so the configuration underlying each displayed number is explicit.

Intent-to-treat accounting. Because three 15 m trials lost the paired beacon's advertisements to vehicle-body blockage and two 25 m trials produced no evaluable fused verdict, every rate is also reported over all 20 initiated trials per separation: BLE-only falls from 16/17 (94.1 %) to 16/20 (80.0 %, CI $[58.4,\ 91.9]$) at 15 m, the hybrid is unchanged at 20/20 (CI $[83.9,\ 100]$) through 15 m because the blockage degrades the BLE-only cell and not the fusion, and the hybrid at 25 m becomes 11/20 (55.0 %, CI $[34.2,\ 74.2]$) against 11/18 (61.1 %, CI $[38.6,\ 79.7]$) per-evaluable. The two exclusion categories are operationally distinct: vehicle-body blockage is a condition a real deployment would also suffer, whereas a missing fused verdict is a measurement artifact of this configuration; the 25 m figure is therefore the one the stricter denominator moves.

Direction. Using two-zone entry order under the $\pm$50 ms clock residual: 56/60 (93.3 %, 95 % CI $[84.1,\ 97.4]$) at 3–10 m and 14/17 (82.4 %, 95 % CI $[59.0,\ 93.8]$) at 15 m; 4 of 77 events (5.2 %) had $|\Delta t| < 1$ s and are reported as undetermined (counted, no direction asserted). Reversal events are classified as undirected and do not enter the direction denominator. Speed is reported as $|v| = 2.355\,\sigma_c/\tau_e$ under the class prior (Section II-D); size and speed are not independently identified (the $\sigma_c/|v|$ degeneracy), and this caveat accompanies every displayed speed.

False confirmations. Zero of 160 empty-scene runs of the two-node configuration produced a false confirmation end-to-end (48,000 s aggregate; rule-of-three 95 % upper bound

1.88 %, i.e., $\leq$ 0.23 false confirmations per hour at 95 % confidence)—a bound on the evaluated quiet controls of the barrier configuration, distinct from the E5 single-device silent sessions. Bidirectional traffic and close-following targets were included in G4. Figure 7 sketches the barrier geometry and the app's own per-separation summary; that summary's 15 m value is the BLE channel and its 25 m value the hybrid, while the measured per-channel rates with confidence intervals are stated above.

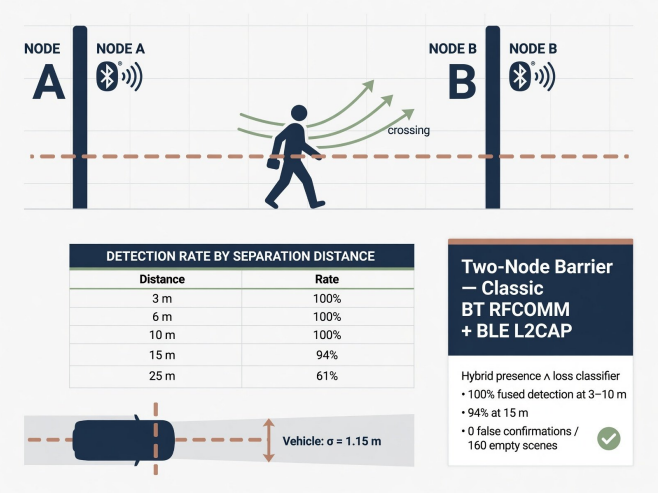


Fig. 7. The *Transit* tab: a two-node barrier over Classic Bluetooth RFCOMM and BLE L2CAP; both a link drop and a link appearance are treated as evidence, and detection is reported at each separation. The rate panel is the app's displayed summary (its 15 m value is the BLE channel, its 25 m value the hybrid); measured per-channel rates with confidence intervals are in Section V-G.

VI. Discussion

The field envelope narrows beyond 12 m: detection falls from 73.3 % at 8 m to 50.0 % at 16 m and 15.0 % at 22 m, while the observed dip depths approach the adaptive trigger floor in the same interval, exactly the behaviour Eq. (1) anticipates when scintillation begins to dominate the shadow term. Two platform effects shape the measured curve. First, Android 15 BLE scan throttling reduces the effective sample rate at distances where the dip margin is thin; the foreground-service configuration mitigates but cannot remove the duty cycle. Second, vehicle transits at road speed compress dip duration toward the detector gate, which explains why vehicle detection sits below pedestrian detection at 8–16 m despite the wider vehicle profile. Both effects are properties of the deployed platform and traffic, not of the radio model, and both argue for reporting distance-conditioned rates rather than a single accuracy figure.

Roles of the mesh and gamification. The mesh turns a bounded single-device sensor into a collective one: cross-node agreement within the stated tolerances confirms events where single-link evidence is weakest, overlapping envelopes tile wider areas, and the reliability bonus steers nodes toward informative geometries. Gamification sustains the participation this collective sensor depends on.

Scope of the validation. This campaign validates the single-device sensing stack—detection, classification, vehicle detection, gated ranging, epoch change detection, silent-control false positives, and the displayed accuracy estimate—together with the two-node transit counter, including direction estimation and end-to-end false confirmations (Section V-G). The cooperative mesh (Section II-D) is exercised here only as a deployed interface; its confirmation and false-confirmation performance requires a distributed multi-node campaign and is left to future work, as is gamification behaviour under sustained multi-user participation.

Threats to validity. (1) Single device, single site: generalisability to other phones and environments is unquantified. (2) Open car park: the multipath environment is simpler than urban canyons; performance may degrade further. (3) Ground-truth timing: manual counters introduce ± 1 s uncertainty, which affects per-event classification more than aggregate counts. (4) Sample size: $n = 15$–20 per cell gives wide confidence intervals at low detection rates; larger batteries of crossings are planned. (5) Weather and traffic were not stratified; sessions span dry daytime conditions only.

VII. Conclusions and Future Work

MeshCrowd demonstrates that commodity smartphones can form a practical, infrastructure-free sensing platform by combining BLE and dual-band Wi-Fi measurements for passive multi-band detection, gated ranging, field-change detection, two-node transit counting, and cooperative mesh sensing, and this paper reports the first complete physical validation of the single-device stack on an unmodified handset. In the field, a single phone detects pedestrian crossings to about 16 m (50 %) and sustains classification above 80 % inside 6 m, detects moving vehicles to 16 m at reduced rate, holds a zero-BLE-false-positive budget over 30 h of silent control, and refuses range estimates exactly where the measured signal cannot support them. The resulting architecture supports applications ranging from **two-node pedestrian and vehicle counting** at entrances and corridors, to **urban crowd and flow monitoring**, **smart-mobility and roadside awareness**, and **emergency situational awareness** where dedicated sensing infrastructure is unavailable, with a gamification layer linking participation and rewards to measurement agreement.

Future evaluation will focus on the distributed multi-device campaign that this campaign deliberately excludes: mesh confirmation and false-confirmation rates across node densities, and gamification dynamics under sustained participation, alongside diverse indoor and outdoor environments, device heterogeneity, placement, multipath, interference, and battery constraints. Mesh resilience against spoofed or adversarial reports also remains untested; future work will address **mesh integrity and incentive robustness**, extending the current agreement-based rewards toward reputation- and quality-aware mechanisms that remain reliable under noisy, spoofed, or coordinated device behavior.

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